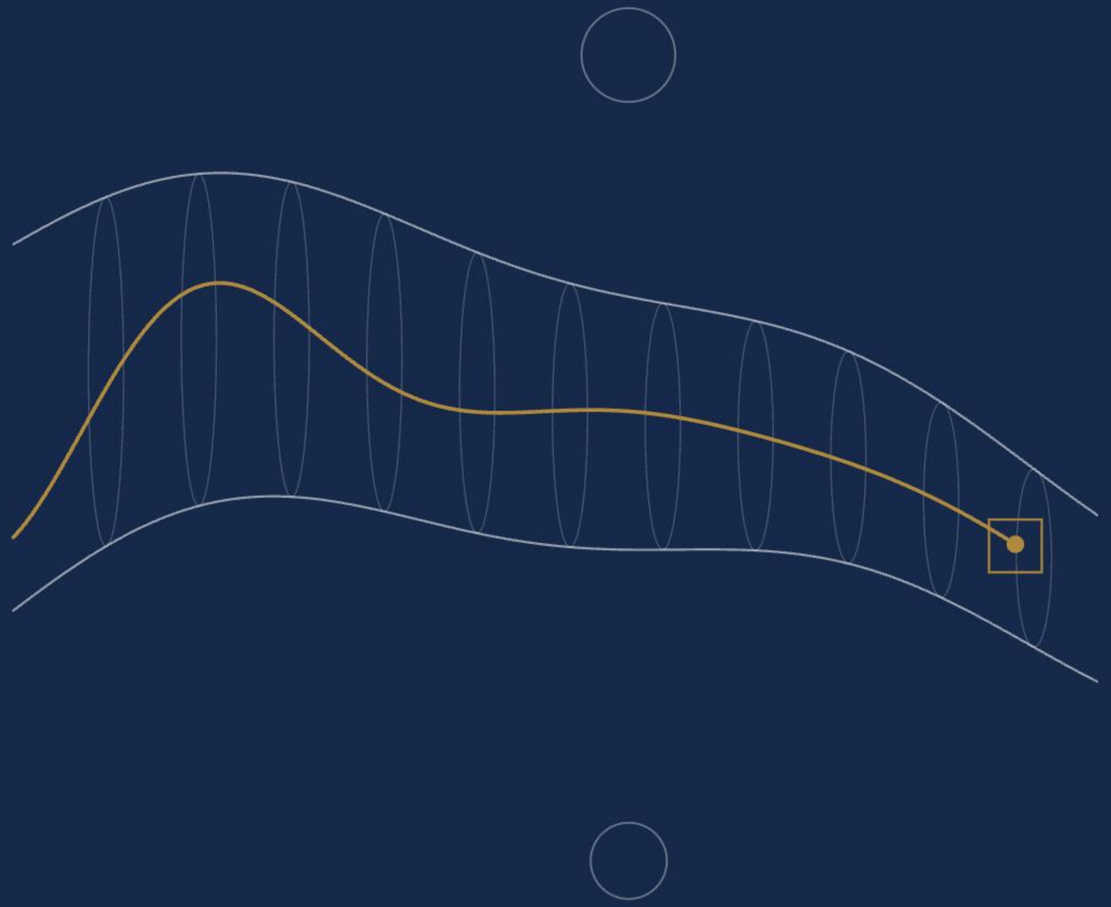
POSTDOCTORAL INTERVIEW · CONTROL & INSTRUMENTATION ENGINEERING · KFUPM

Safe, Compliant and Adaptive Control for Robots that Interact



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Robert Bosch Centre for Cyber-Physical Systems, IISc Bangalore

4 November 2026

PROFILE

Background

CURRENT

Research Associate, FOCAS Lab

Robert Bosch Centre for Cyber-Physical Systems, IISc Bangalore

With Dr. Pushpak Jagtap: spatiotemporal tubes and funnel-based interaction control

PHD, 2026

Electrical Engineering, IIT Delhi

Safe and Compliant Adaptive Control Strategies for Human–Robot Interaction

Advisor: Prof. Indra Narayan Kar

RESEARCH QUESTION

How can a robot yield to people while provably staying safe?

- 01 **Compliant interaction**
- 02 **Hard safety constraints**
- 03 **Limited model knowledge**
- 04 **Explicit time guarantees**

3 journal articles (IJSS ×2, IJHCI)

6 under review (4 journal + 2 conference)

2 conference papers (IEEE CoDIT, IFAC ACODS)

2 controllers validated on hardware

RESEARCH THEME

Interaction that is safe by construction

01
Learn

Identify unknown parameters **in fixed time**, from initial rather than persistent excitation.

Certificate: explicit settling-time bound $\mathcal{T}_{\max}$

02
Constrain

Never leave the safe set — **even while yielding** to the human.

Certificate: forward invariance; recursive feasibility

03
Estimate

Reconstruct **unmeasured states**, treating the human force as an unknown input.

Certificate: contraction ⇒ exponential convergence

Separate what must be guaranteed from what can be learned — and prove both.

MOTIVATION

Compliance and safety under uncertainty

INTUITION: WHAT IMPEDANCE CONTROL DOES

Stiff position control

robot holds the path: the push meets a wall



Impedance control

robot behaves like a spring–damper: yields, then returns







	Model-free	Parameter convergence	Explicit time	Hard safety
Classical impedance	×	–	–	×
Adaptive impedance	✓	PE only	×	×
Invariance control	×	–	–	✓
CBF / MPC based	×	–	–	✓
BLF / PPC admittance	✓	×	×	✓†
This work	✓	✓	✓	✓

† conservative: bounds tied to the barrier construction rather than the task. “Model-free”: unknown parameters with known regressor structure.

TASK-SPACE DYNAMICS AND TARGET IMPEDANCE

$$M_p(q)\ddot{p} + C_p(q,\dot{q})\dot{p} + G_p(q) = J^{\dagger\top}(q)\,\tau + f_{ext}$$

$$M\ddot{e} + D\dot{e} + Ke = f_{ext}, \qquad e = p - p_d$$

The gap. No single framework delivered compliance, parameter convergence in bounded time and hard state constraints — together, under model uncertainty and disturbances.

Fixed-time adaptive impedance control

DESIGN

$$M(\dot{v} + \lambda v) = M\ddot{e} + D\dot{e} + Ke - f_{ext}$$

$$\tau = J^{\top}\left(\Phi\hat{\Theta} - f_{ext} - k_1\lceil v \rfloor^{1-\alpha} - k_2\lceil v \rfloor^{1+\alpha}\right)$$

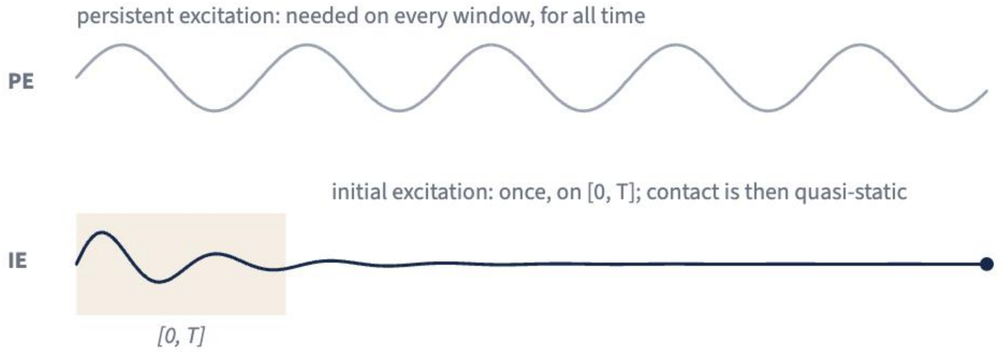
$$\dot{\hat{\Theta}} = \Gamma\Phi^{\top}v + \Gamma\sum_{j\in\{F,IF\}}\left(\lceil \varepsilon_j \rfloor^{1-\alpha} + \lceil \varepsilon_j \rfloor^{1+\alpha}\right)$$

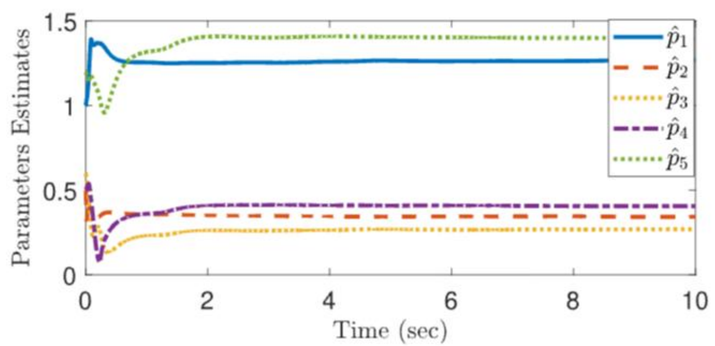
$\lceil x \rfloor^{a} = |x|^{a}\,\mathrm{sign}(x)$ elementwise, $\alpha \in (0,1)$; ε_j: prediction errors of filtered regressors.

Theorem. If the filtered regressor is initially exciting on $[0, \epsilon]$, then for every initial condition

$$v(t) = 0, \quad \tilde{\Theta}(t) = 0 \qquad \forall\, t \geq \mathcal{T}_{\max} = \epsilon + \frac{\pi}{\alpha\sqrt{\gamma_3\gamma_4}}.$$

Submitted to the European Journal of Control — with D. Efimov and R. Ushirobira (Inria)





Two-link simulation: all five estimates settle well inside $\mathcal{T}_{\max} = 12.3$ s.
Hardware (Quanser SRV02): $\mathcal{T}_{\max} = 39.7$ s, smooth and bounded input.

01 Initial, not persistent, excitation

Excitation is needed only on $[0, \epsilon]$ — realistic for contact tasks that turn quasi-static.

02 A deadline known in advance

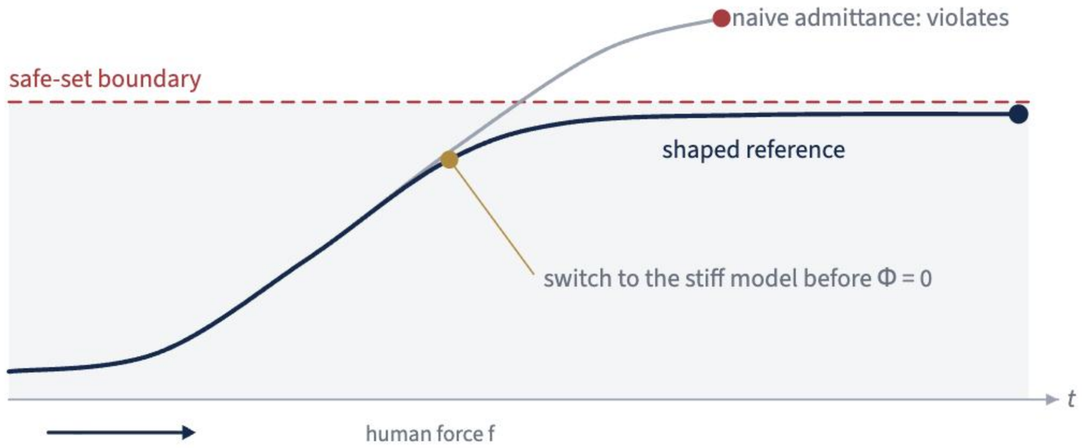
$\mathcal{T}_{\max}$ is set by the design gains and the excitation level, not by the initial conditions.

03 Smooth, and tested on hardware

Continuous control law, no chattering; on the Quanser SRV02 the estimates settle within $\mathcal{T}_{\max} = 39.7$ s.

Safety in admittance control by reference shaping

INTUITION: THE SAME PUSH, TWO REFERENCES

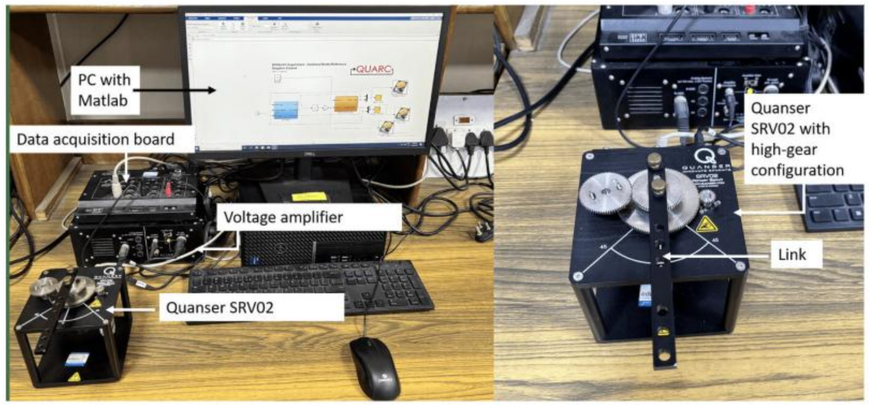


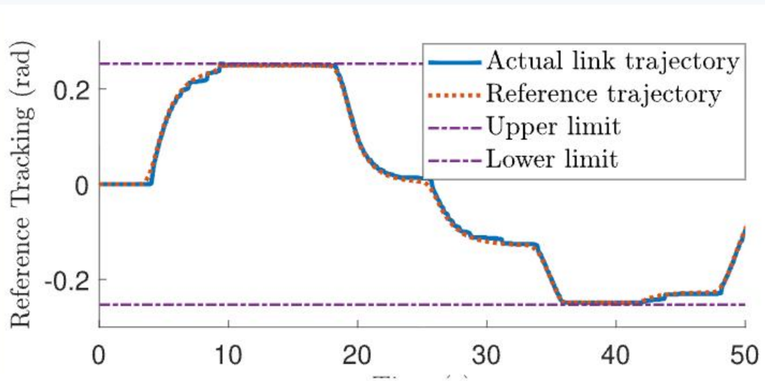
THEORY: SWITCHED REFERENCE MODEL AND INVARIANCE FUNCTION

$$\dot{E}_r = A_\sigma E_r + B_a f_{ext}, \qquad \sigma \in \{\text{compliant, stiff}\}$$

$$\Phi(E_r) = \begin{cases} h, & \dot{h} \le 0 \\ h + \dfrac{\dot{h}^2}{2\gamma}, & \text{otherwise} \end{cases} \qquad \mathcal{G}_1 \oplus \mathcal{D} \subseteq \mathcal{S}_r$$

Safe set $\{h \le 0\}$; $\Phi \le 0 \Leftrightarrow$ the braking distance at deceleration γ fits before the boundary. $\mathcal{G}_1 = \{\Phi \le 0\}$ invariant; MRAC error bound $\mathcal{D}$.



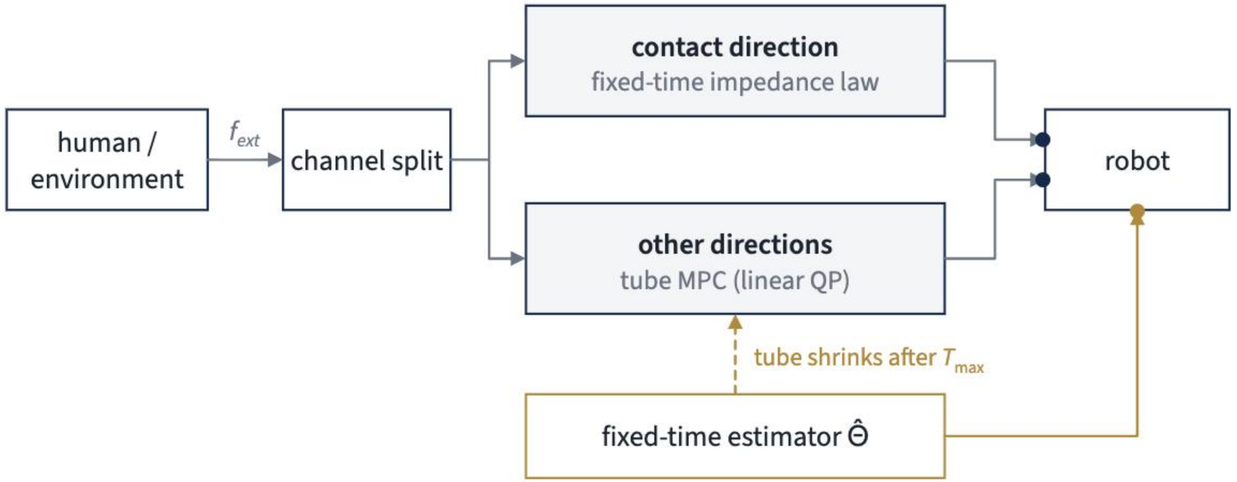


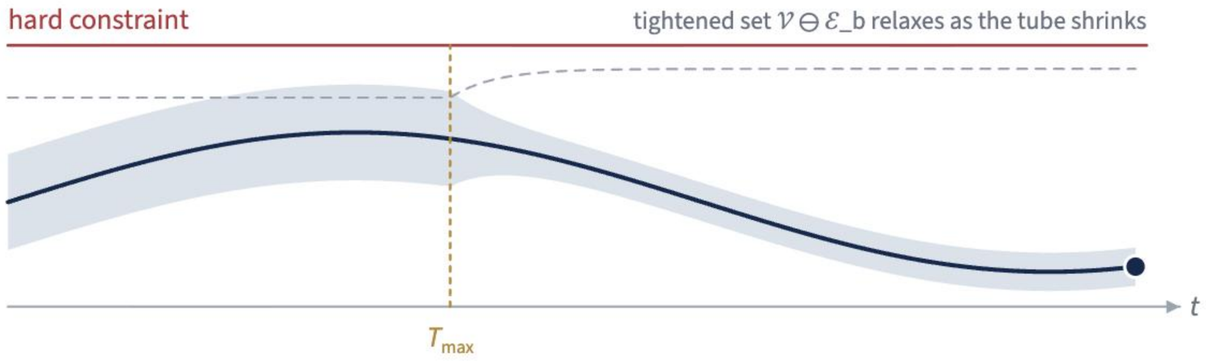
HARDWARE AND RESULT

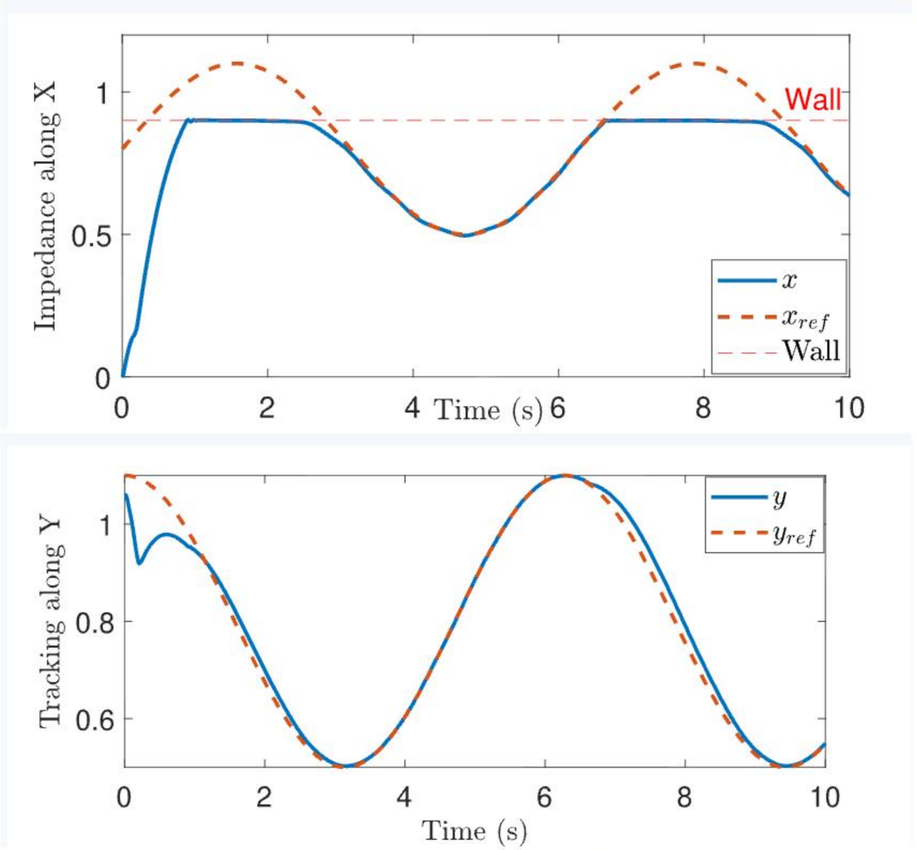
The operator pushes the link both ways ($|f_{ext}| \lesssim 2$ N): the link **complies** but **never crosses the limits**. No barrier terms in the control law.

Int. J. Human–Computer Interaction (2026) · IEEE CoDIT (2024)

Fixed-time adaptation with tube MPC



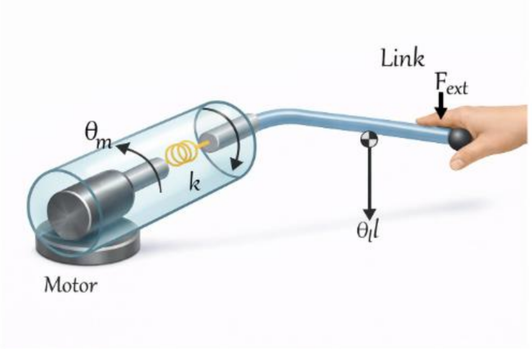




Simultaneously: compliant contact with a wall along X (top) and accurate tracking along Y, with the Y-channel constraint enforced (bottom).

In revision, IEEE Transactions on Automation Science and Engineering

Unknown-input dynamic observer via contraction



Flexible joints give inherent compliance, but link-side states are hard to measure and the human force is unknown.

SYSTEM AND ANNIHILATOR

$$\dot{x} = Ax + g(y,u) + \phi(x) + Fd, \quad y = Cx$$

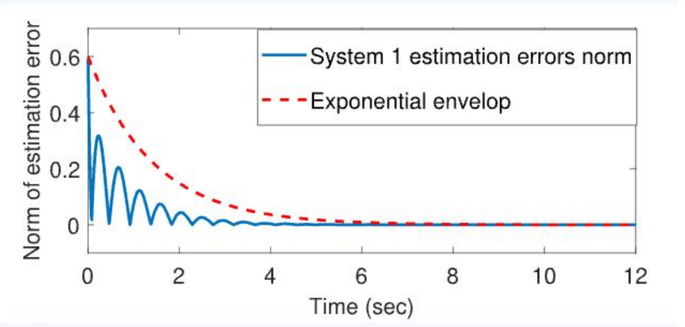
$$TF = 0 \;\Rightarrow\; T\dot{x} = TAx + Tg + T\phi(x)$$

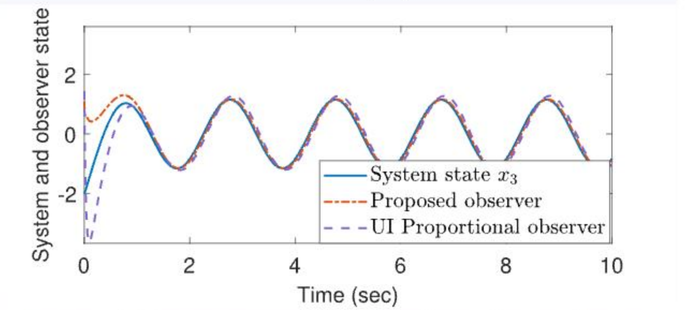
DYNAMIC OBSERVER

$$\dot{z} = Pz + Qy + Tg + T\phi(\hat{x}) + Rv$$

$$\dot{v} = Xz + Yy + Sv, \qquad \hat{x} = Wz + Zy$$

Exists iff $\operatorname{rank}\begin{bmatrix} T \\ C \end{bmatrix} = n$. Contraction of a virtual system, certified by an LMI, gives exponential convergence for any initial estimate.





Error within an exponential envelope; faster than a proportional UI observer (simulation).

Int. J. Systems Science (2025)

01 No force sensor needed

The human force is treated as an unknown input, so link-side states are recovered without measuring it.

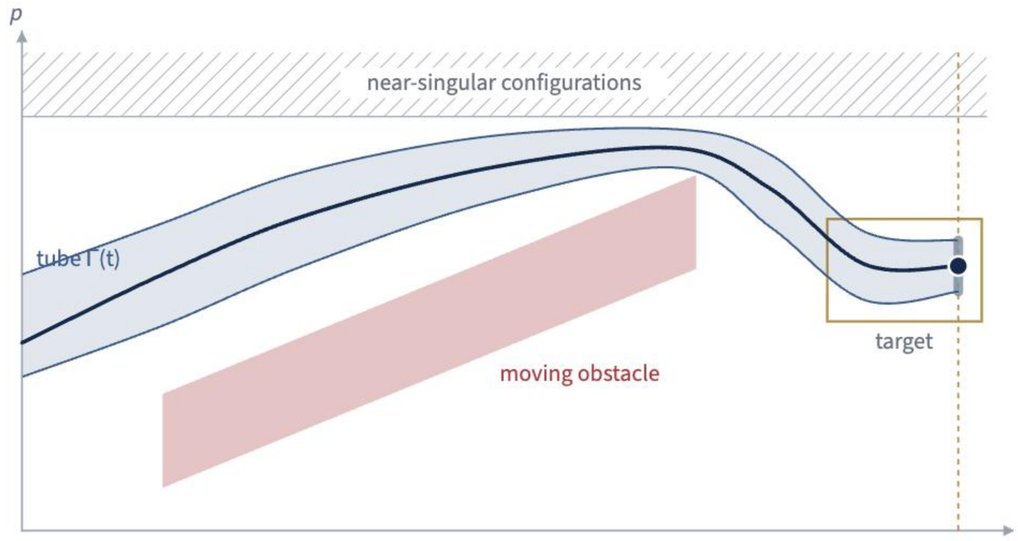
02 More design freedom

Extra observer dynamics (X, Y, S) enlarge the LMI feasible set; proportional and PI observers are special cases.

03 Converges from any guess

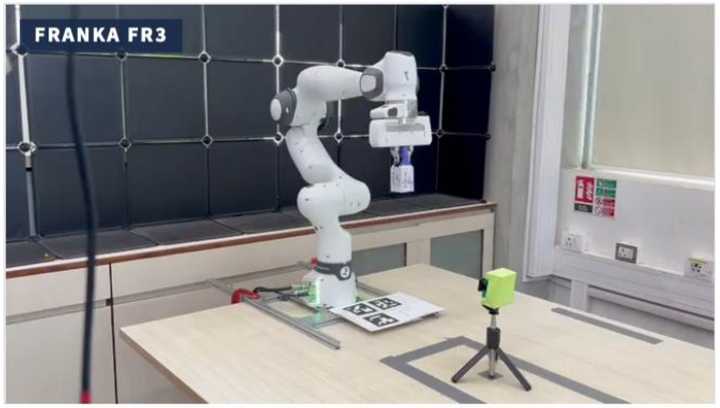
Contraction of a virtual system gives exponential convergence for every initial estimate.

Spatiotemporal tubes for safe task-space control





hatched: singular configurations (arm stretched / folded)



Tracking a target moved by hand (4× speed).

1 Draw a corridor in space and time

A tube $\Gamma(t) = \{p : \gamma_L(t) < p < \gamma_U(t)\}$ says where the end effector may be at every instant.

2 Bend it around danger

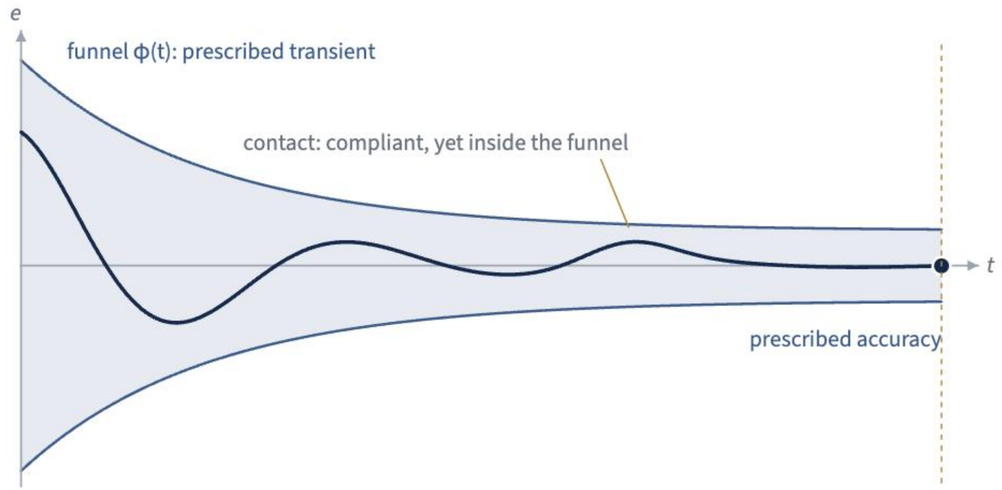
The tube avoids moving obstacles $\mathcal{O}(t)$ and singular arm configurations, and reaches the target on time.

3 Stay inside, in closed form

An approximation-free law keeps $p(t) \in \Gamma(t)$ for all t: no model learning, no online optimisation.

Safety is decided when the tube is drawn; the controller only has to stay inside it.

Funnel-based interaction control



IDEA

Keep the interaction error inside a designed funnel, $\|e(t)\| < \varphi(t)$, with a low-complexity, model-free law

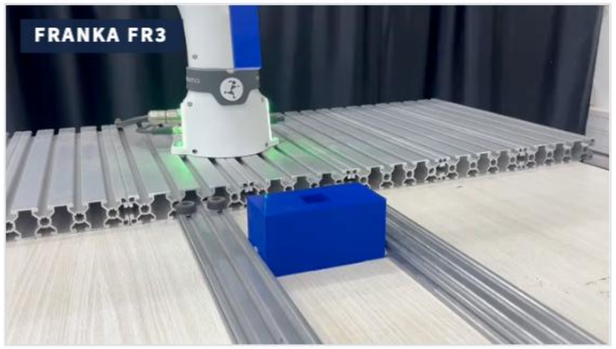
$$u = -\frac{k}{1 - \|e\|^2/\varphi(t)^2}\, e$$

Performance by design; no parameter estimation; the gain rises only near the funnel boundary.

WHY FUNNELS ARE HARD IN PHYSICAL INTERACTION

- **An unknown push** drives the error towards the boundary, where the gain grows sharply: torque peaks and saturation.
- **Noise and delay** near the boundary are amplified by the high gain.
- **Compliance vs. a tight funnel:** a narrow funnel makes the robot stiff to the human; a wide one weakens the guarantee.
- **Interaction stability** with an unknown human is not given by the funnel alone.

Also in progress: observer-based fixed-time nested sliding-mode control of a flexible-link manipulator (with IIT Delhi).



Contact-rich peg insertion (3× speed).

> The same tube and funnel geometry gives **safety** (spatiotemporal tubes) and **prescribed interaction performance** (funnels); fixed-time tools give the **timing certificates**.

TRACK RECORD

Publications

JOURNAL ARTICLES

C. K. Paul, B. S. Dey, I. N. Kar. Safety in admittance control using reference trajectory shaping. *Int. J. Human–Computer Interaction*, 2026.	PUBLISHED
B. S. Dey, **C. K. Paul**, I. N. Kar. A matrix measure framework for incremental stability of mixed-mode switched systems. *Int. J. Systems Science*, 2026.	PUBLISHED
C. K. Paul, I. N. Kar, S. Janardhanan. Unknown input dynamic observer for a class of nonlinear systems using contraction analysis. *Int. J. Systems Science*, 2025.	PUBLISHED

UNDER REVIEW: JOURNALS

C. K. Paul, B. S. Dey, I. N. Kar. Safe physical human–robot interaction via fixed-time concurrent-learning impedance control. *IEEE T-ASE*.	IN REVISION
C. K. Paul, K. Nath, I. N. Kar, D. Efimov, R. Ushirobira. Fixed-time adaptive impedance control with excitation-aware parameter estimation. *European Journal of Control*.	SUBMITTED
C. K. Paul, K. Nath, I. N. Kar, D. Efimov, R. Ushirobira. Fixed-time model reference adaptive control via excitation-preserving delayed regressors. *Automatica*.	SUBMITTED
A. Sharma, **C. K. Paul**, S. Janardhanan. Observer-based fixed-time nested sliding-mode control for tip-position regulation of a single-link flexible manipulator. *Trans. Inst. Measurement and Control*.	SUBMITTED

CONFERENCES

C. K. Paul, B. S. Dey, I. N. Kar. Safe human–robot interaction using switched model reference admittance control. *IEEE CoDIT*, Malta, 2024.	PUBLISHED
C. K. Paul, N. Patel, B. S. Dey, I. N. Kar. Adaptive impedance control: a contraction theory approach. *IFAC ACODS*, New Delhi, 2024 (IFAC-PapersOnLine 57).	PUBLISHED
Two manuscripts under double-blind review (IEEE RA-L, ICRA).	UNDER REVIEW

METHODS

Fixed-time stability, adaptive and concurrent learning, set invariance, tube MPC, contraction, sliding modes, funnels and tubes.

IMPLEMENTATION

Real-time experiments on the **Franka Research 3 (FR3)** and the Quanser SRV02; ROS 2, MATLAB/Simulink, Python.

TEACHING

IIT Delhi (2022–25): labs and tutorials in basic electrical, control engineering, nonlinear systems and robotics. NIT Jalandhar (2025): discrete-time control. DSEU (2022–24): basic electrical and electronics.

COLLABORATION

IIT Delhi, IISc Bangalore, Inria (France).

OUTLOOK

Research plan at KFUPM CIE

T1
Safe human–robot collaboration

- Tube and funnel interaction control with fixed-time adaptation on cobot arms
- Time-varying, unmodelled human dynamics

T2
Observer-based and fault-tolerant safety

- Safe interaction without force sensing or full state
- Tubes that survive actuator faults; discrete-time guarantees

T3
Safe multi-robot and aerial systems

- Tubes for fleets and formations under communication limits
- Aerial manipulation; links with CBF-based task allocation

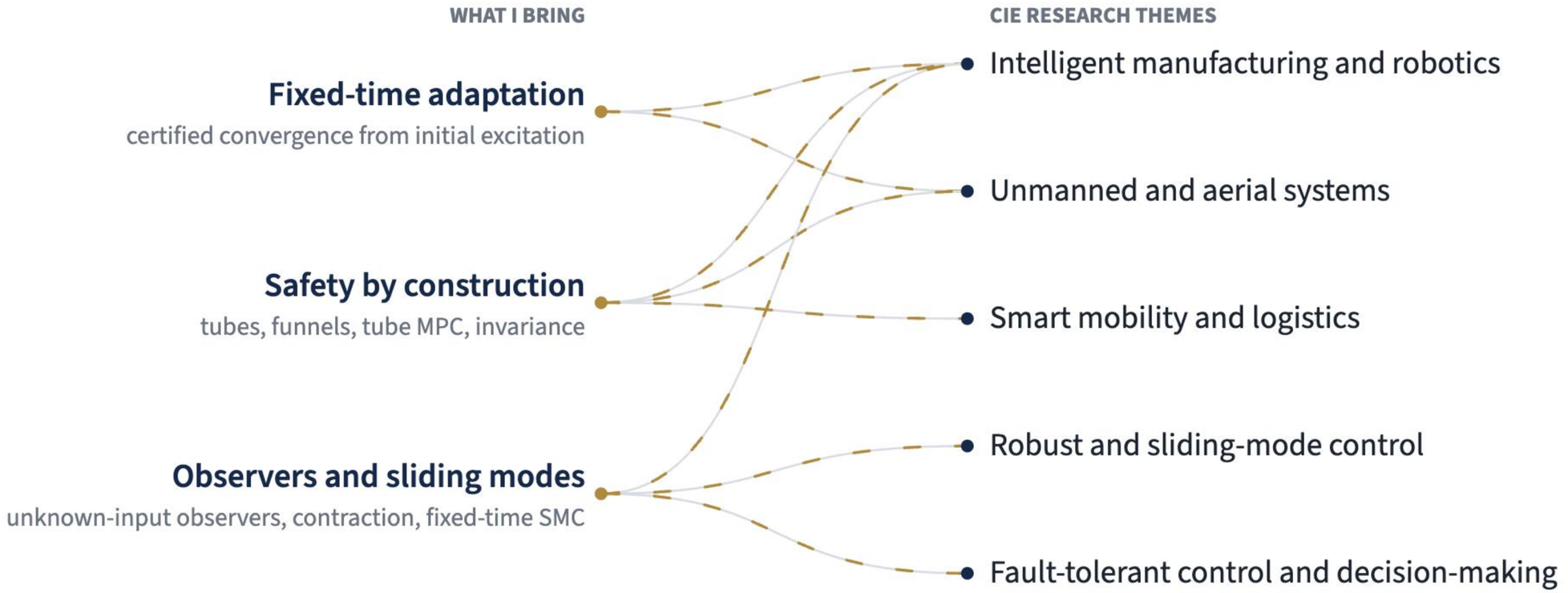
Months 0–8
T1 on lab cobots; T2 theory; first journal submission

Months 8–16
T1 experiments; T3 in simulation; conference papers

Months 16–24
Integration and demonstrations; joint proposals; student co-supervision

OUTLOOK

Fit with the department



> I would bring certified safety and fixed-time learning for physical interaction, and connect them to the department's robotics, unmanned-systems and logistics projects.

Thank you

- Compliant, provably safe interaction control with explicit settling-time certificates
- Hardware-validated on a one-degree-of-freedom testbed; published and under review in leading journals
- A 24-month plan on certified-safe autonomy around people, aligned with CIE research



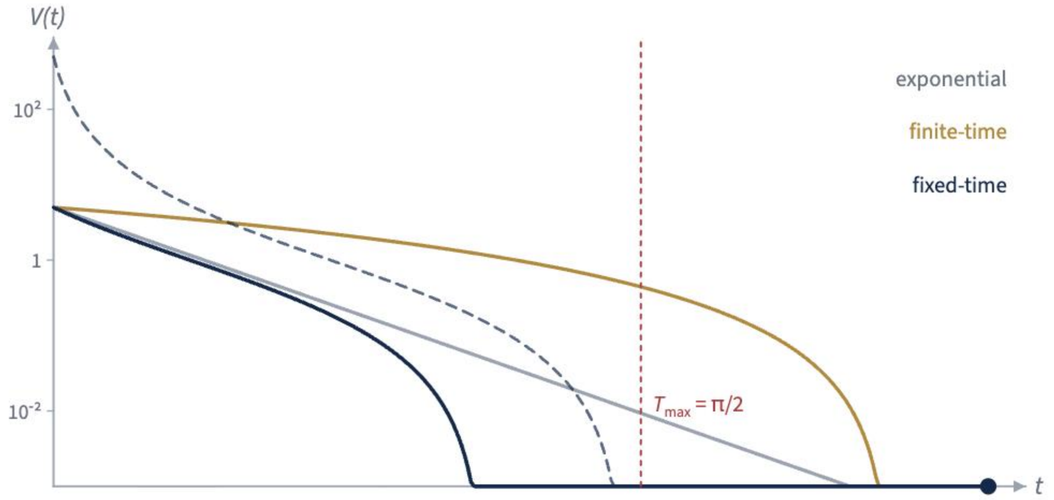
PAPERS, VIDEOS, CODE AND CV

www.ckpaul.in

Chayan Kumar Paul · chayankumar1@iisc.ac.in | Backup slides: press ↓

BACKUP A

Stability notions and the fixed-time bound



$\dot V = -4V\cdot\dot V = -2\sqrt{V}\cdot\dot V = -2\sqrt{V} - 2V^{3/2}$ (log scale). Dashed: fixed-time from $V_0 = 500$, still settled before $\mathcal{T}_{\max}$.

LEMMA (POLYAKOV, IEEE TAC 2012)

If $\dot{\mathcal{V}} \le -\gamma_a \mathcal{V}^{1+p} - \gamma_b \mathcal{V}^{1-p}$ with $0 < p < 1$, the origin is fixed-time stable and

$$\mathcal{T}(x_0) \le \mathcal{T}_{\max} = \frac{\pi}{2p\sqrt{\gamma_a \gamma_b}}.$$

Proof idea: $w = \mathcal{V}^p$ gives $\dot w \le -p(\gamma_b + \gamma_a w^2)$; the arctangent is bounded by $\pi/2$.

IN MY DESIGNS

$\lceil\cdot\rfloor^{1-\alpha}$ acts near the origin, $\lceil\cdot\rfloor^{1+\alpha}$ far away. The right-hand side is continuous, so there is no sliding-mode chattering. A smaller $\mathcal{T}_{\max}$ requires larger Γ, k_1, k_2 and gives higher torque peaks.

BACKUP B

Why initial excitation is enough

REGRESSOR FILTERS (ZERO INITIAL STATES)

$$Z\Theta = \tau_1$$

$$\dot{Z}_F = -kZ_F + Z, \quad \dot{\tau}_{1F} = -k\tau_{1F} + \tau_1$$

$$\dot{Z}_{IF} = -kZ_{IF} + Z_F^\top Z_F$$

If IE holds on $[0, \epsilon]$, then $\lambda_{\min}(Z_{IF}) \geq \mu > 0$ on $[\epsilon, \epsilon_1]$ — long enough to converge before $\mathcal{T}_{\max} \leq \epsilon_1$, after which $\tilde{\Theta} = 0$ is invariant (disturbance-free case). With forgetting $k > 0$ and no new excitation, $\lambda_{\min}(Z_{IF})$ eventually decays; hence the window.

PHYSICAL VIEW

A contact task has a short transient followed by quasi-static contact: PE fails, IE holds.

PRACTICAL CHECK

Monitor $\lambda_{\min}(Z_{IF}(t))$ online; the certificate is active once it exceeds μ.

SPECIAL CASE

$\alpha = 0$ recovers the classical asymptotic combined / concurrent-learning update.

BACKUP C

Tube MPC: formulation, feasibility, computation

TUBE-MPC PROBLEM

$$\min_{\bar{u}} \ \sum_{j=0}^{N-1} \left(\|\bar{v}_{k+j}\|_Q^2 + \|\bar{u}_{k+j}\|_R^2 \right) + \|\bar{v}_{k+N}\|_P^2$$

$$\text{s.t. } \bar{v}_{k+j+1} = A_d \bar{v}_{k+j} + B_d \bar{u}_{k+j},$$

$$\bar{v}_{k+j} \in \mathcal{V} \ominus \mathcal{E}_b, \quad \bar{v}_{k+N} \in \mathcal{V}_f,$$

$$u_k = \bar{u}_k^* + K_d (v_k - \bar{v}_k), \quad A_d + B_d K_d \text{ Schur}$$

GUARANTEES

Recursive feasibility via the shifted sequence; J^* is a Lyapunov function, so $\bar{v}_k \to 0$; the true state stays in $\bar{v}_k \oplus \mathcal{E}_b$, with $\mathcal{E}_b$ robust positively invariant.

LESS CONSERVATIVE

The disturbance set has a parametric part that vanishes after $\mathcal{T}_{\max}$, so the tube can be recomputed smaller.

COMPUTATION

$A_d = e^{-\lambda T_s} I$ with box constraints gives a small dense QP ($N = 40$, $T_s = 1$ ms). Terminal ingredients are computed offline.

IF THE BUDGET IS TIGHT

Explicit (multi-parametric) MPC, warm-started first-order solvers, or a shorter horizon with a larger terminal set.

BACKUP D

Contraction and the observer design equations

CONTRACTION

$\dot{x} = f(x,t)$ is contracting in a metric $\mathcal{M}(x,t) \succ 0$ if

$$\frac{\partial f}{\partial x}^{\top} \mathcal{M} + \dot{\mathcal{M}} + \mathcal{M}\frac{\partial f}{\partial x} \preceq -2\beta\mathcal{M}, \quad \beta > 0.$$

All trajectories converge to each other exponentially. A virtual system with particular solutions x and $\hat{x}$ then gives $\hat{x} \to x$.

LINEAR MATRIX EQUATIONS

$$PT - TA + QC = 0, \quad XT + YC = 0, \quad WT + ZC = I$$

Stack as $\Pi\Sigma = \Lambda$ with $\Pi = [P\ Q\ X\ Y\ W\ Z]$; solvable iff $\operatorname{rank}\begin{bmatrix}\Sigma \\ \Lambda\end{bmatrix} = \operatorname{rank}\Sigma$; general solution $\Pi = \Pi_0 + \mathcal{Z}_1(I - \Sigma\Sigma^{\dagger})$. The free block $\mathcal{Z}_1$ is used to satisfy the convergence LMI at the vertices bounding the Jacobian of ϕ.

BACKUP E

Safety tools: how they compare

Method	Model needed	Online cost	Constraint type	Typical caveat
CBF–QP	control-affine model	QP each step	state set $h(x) \geq 0$	feasibility under input limits; high relative degree
Barrier Lyapunov function	partial (backstepping)	closed form	error / state bounds	initial-condition requirements; design complexity
Prescribed performance / funnel	structural only	closed form	error envelope $\varphi(t)$	gain grows near the boundary; noise sensitivity
Spatiotemporal tubes	structural only	closed form	time-varying state tube	the tube must be designed (obstacle knowledge)
Tube MPC	nominal linear model	QP each step	state and input sets	computation; conservatism of the invariant set
Reference shaping (mine)	none in the safety layer	closed form	state set via Φ	needs a good tracking loop; switching design

BACKUP F

Assumptions and limitations

STANDING ASSUMPTIONS

- Dynamics linear in the parameters (known regressor)
- Interaction force measured; disturbances bounded with a known bound
- Full state available in Ch. 3–5; safe set known, initial state inside it
- Non-singular Jacobian along the task (now addressed with tubes)

LIMITATIONS AND NEXT STEPS

- Hardware so far is single-DoF → multi-DoF cobot experiments
- Ch. 5–6 validated in simulation → hardware
- $\mathcal{T}_{\max}$ is a worst-case bound → tighter estimates
- Continuous-time proofs → discrete-time guarantees
- Observer and controller designed separately → observer-based safe interaction